Supplementary Material for: Accurate but Non-Compliant: Auditing the Administrative-Label Convention for AKI Risk Modelling across 206 Hospitals

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Note. Plain section numbers (e.g. Section 5.3) refer to the main paper; sections, tables and figures of this document carry an S prefix.

S1 Pre-Registration, Deviations, and Undelivered Items

The protocol's claim is that pre-registered pass criteria make an audit falsifiable, and that claim is only as good as the completeness with which departures are declared. Table S2 lists every deviation known to us between the ratified pre-registration (`prereg-v1.3`), the released artifacts, and the results reported here, including the unflattering ones. Rows marked NOT DELIVERED are registered items we did not run; rows marked CONTROL FAILED are negative controls that did not fire on their target axis, which per §6 of the pre-registration counts *against* the protocol. No pass *threshold* was ever altered.

Axis	Artifact	Pass criterion
T1 replay	Decision trace: ids, version and config hashes, sub-symbolic output, constraint evaluations	0 prediction mismatches on re-execution *and* matching model-version hash
T2 rule	Per-decision soft/hard constraint log	0 hard-constraint violations; soft firings logged
T3 fallback	Calibration and abstention policy	ECE below a pre-declared bound; abstention no lower under shift

Table S1: The three audit axes. Each is a test whose pass criterion is fixed before test-fold evaluation. Verdicts are reported separately, never collapsed into one score.

Three points deserve emphasis rather than burial. First, a pre-registration that quietly drops its unrun arms provides weaker evidence than one that reports them missing, so rows 4 and 6 are named as undelivered; row 5, the pure-neural baseline, is DELIVERED IN V1.4 and reported in Section 5 and Appendix S6, with the caveat that its accuracy equality with the audited model is by construction and carries no statistical content. Second, the abstention threshold τ was registered as selected once and frozen with the value recorded in a lock file; the released code re-selects it on the validation fold within each seed, yielding 19 distinct values over 20 seeds in the range 0.6441–0.6587, and no lock file exists. Selection never touches the test fold, and every verdict is stable across that spread, but the registered artifact was not produced. Third, the subgroup floor of $n \geq 500$ (pre-registration §2) was declared in v1.0 and first enforced in code in this revision; enforcing it renders the Black-patient model-dependent comparison UNDETERMINED (Section 5.3).

Severity sensitivity was pre-declared. The severity-definition sensitivity arm reported in Section 5 and Appendix S5 was declared in the pre-registration *before* it was run, and is a sensitivity arm only: the v1.3 primary specification (the absolute limb) remains primary and is unchanged.

Test suite. Of the broader released suite, 16 tests are tied directly to the headline audit: five inherited tests of the calibration and bootstrap metrics, and 11 added here to cover the components that produce the headline claim. Of those 11, five call `audit.run_T2` directly on synthetic fixtures and pin its semantics: that only confident negatives on severe cases count, that a deferral is not a violation, that the enforced policy reaches zero violations by construction, that a model emitting no evidence is reported as such rather than as compliant, and that the violation rate is taken over deployed rather than all severe decisions. Three call the real `experiment._aggregate` with synthetic per-seed records in the shape the pipeline produces, and would fail if the AND-condition or the tail quantiles of the certificate changed. The remaining three concern the severe-case mask, and we scope them precisely: two are *integration* checks that load the locally built cohort artifact and SKIP — rather than pass vacuously — when it is absent, while the third is a data-free guard that reads `cohort.py` and fails on a silent rename of the severe column or threshold key, so it always runs. We do not claim data-free unit coverage of cohort construction, whose severe-mask logic is inline in `cohort.build_cohort` and requires credentialed source files.

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#	Registered item (§ numbers refer to the pre-registration, not to this paper)	Deviation	Status	Consequence / where reported
1	§3 T2 applicability	Absolute creatinine $\geq 4.0 \rightarrow$ acute-severe (KDIGO rise)	CORRECTED	Clinical validity (CKD confound); Sec. 4
2	§3 T2 window	7-day peak $\rightarrow$ first-24h decision window	CORRECTED	Temporal leakage removed; Sec. 4
3	§3 T2 certificate	Compliance–coverage certificate added (v1.3)	TIGHTENED	Closes abstention-gaming; Sec. 5.4
4	§1 Task A (ICU in-hospital mortality)	Not run; no result reported	NOT DELIVERED	One clinical task across two databases, not two tasks
5	§7.4 pure-neural baseline	Registered, previously unexecuted; run in v1.4, reported in prose and carried in Table S12 as a "not computable" row	DELIVERED V1.4	AUROC equality is by construction (the class wraps the audited model), not an empirical result; Sec. 5
6	§7.3 SPL arm (DeepProbLog fallback)	Neither the SPL arm nor its registered fallback was run	NOT DELIVERED	Two of three registered integration styles evaluated
7	§3 T3 abstention threshold τ	Registered as frozen in a lock file; implemented as re-selected on the validation fold per seed (19 values, 0.6441–0.6587); no lock file exists	DEVIATION, REPORTED	No test-fold leakage and verdicts stable across the spread, but the registered artifact was not produced
8	§6 NC-A (eICU)	Did not flag its target axis (ECE CI upper $0.088 < 0.10$)	CONTROL FAILED	T3 unvalidated on eICU; Sec. 6

Table S2: Complete deviations log against `prereg-v1.3`, part 1 of 3 (rows 1–8; part 2 in Table S3). Rows 1–3 predate the v1.3 freeze. No pass threshold was ever altered.

#	Registered item (§ numbers refer to the pre-registration, not to this paper)	Deviation	Status	Consequence / where reported
9	§6 NC-C (both databases)	Did not flag T3-shift in any released run	CONTROL FAILED	T3-shift sub-criterion excluded from every reported verdict
10	§2 split stratification	Described as stratified; implemented as grouped (`GroupShuffleSplit`) without stratification	DESCRIPTION FIXED	Immaterial at $n \approx$ 113k and 18.3% prevalence
11	§3 T3 ECE binning	10 equal-width bins declared; only 5 reachable under top-label confidence	DISCLOSED	Biases ECE towards the T3 pass; Sec. 3
12	§2 `min_subgroup_n: 500`	Declared in v1.0, first enforced in code in v1.4	ENFORCED V1.4	Renders the Black-patient model-dependent comparison UNDETERMINED; Sec. 5.3
13	(not registered)	Trajectory-stratified and directly standardised label validity, plus a within-window chronicity proxy and a logistic robustness check, added in v1.5 after the crude subgroup gap was known	EXPLORATORY, POST HOC	Tests whether measured case mix explains the gap; *not* adjusted for pre-existing CKD, comorbidity or site; Sec. 5.3
14	(not registered)	Matched positive control on the cohort intersection (same rows, features, splits and seeds; target varied), added in v1.5	EXPLORATORY, POST HOC	Removes the cohort-composition objection to the registered positive control; Sec. 5.5
15	(not registered)	Prior-CKD sensitivity analysis: `10_prior_ckd_sensitivity.py` stratifies on CKD documented in an admission discharged *before* the audited one, excluding the audited admission's own codes	EXPLORATORY, POST HOC; RUN	Released unrun in a prior revision, then run once the credentialed tables became available. Blocks A–E ran *as released* (diff: 97 lines added, none removed) and narrow our own subgroup claim: the gradient attenuates and the adjusted race coefficient no longer excludes zero. Block G was *added after* block E revealed unanticipated renal-replacement prevalence, and its interpretation was rewritten after its first output contradicted our initial reading; it is post-hoc analysis chosen after seeing results (agreement 0.897 without prior renal disease against 0.079 with prior dialysis or transplant). Selection-limited: prior history is itself differentially observable (82.9% vs 69.7%); Appendix S7

Table S3: Complete deviations log, part 2 of 3 (rows 9–15; part 1 in Table S2; part 3 in Table S4). Rows 13–14 are exploratory analyses added after the results they qualify were known; row 15 is the analysis we could not run, specified so a credentialed team can. No pass threshold was ever altered.

#	Registered item (§ numbers refer to the pre-registration, not to this paper)	Deviation	Status	Consequence / where reported
16	§4 positive control	Effect ceiling computed for the matched positive control: the KDIGO target is already positive on 95.4% of constraint-applicable cases, so 81% of the violation-rate drop follows from target base rates	DISCLOSED	Reported so the control is not read as clean causal evidence. We decline to promote it to a primary causal claim on this basis; Sec. 5.5
17	§2 constraint validity	Clinician adjudication of the constraint: `11_clinician_adjudication.py`, blinded stratified 200-case packet, falsification thresholds fixed in source before any adjudication exists	RUN (V1.12); ONE ADJUDICATOR	162/200 decided, 38 blank; both pre-declared thresholds pass (cell B 63/63, κ 0.835 vs rule, 0.023 vs code); cross-tabulation added in v1.13. Validates the operationalisation against a nephrologist's reading of KDIGO, not clinical AKI independently; Sec. 5.6 and Section S14

Table S4: Complete deviations log, part 3 of 3 (rows 16–17). These additions disclose post-registration analyses and the clinician-adjudication protocol, now run. The post-submission entries (v1.12: restricted-cohort model certificate, held-out-hospital certificate, adjudication; v1.13: cross-tabulation and clustered CIs) are logged in `preregistration_v14.md` and tabulated in Section S14. No pass threshold was ever altered.

S2 Pima Demonstration: A Model That Passes

The protocol should be able to pass a reasonable model, not only fail one. We demonstrate this on the UCI Pima Indians Diabetes dataset (Smith et al. 1988) (768 records, 268 positive; physiologically impossible zeros in glucose, blood pressure, skin thickness, insulin and BMI treated as missing and imputed with *train-split* medians), with a 70/30 stratified split (537 train, 231 test) and a probabilistic linear core on standardised features, run without a post-hoc calibration wrapper — unlike the MIMIC-IV arm, which is isotonic-calibrated. This matters for reading the T3 result below: the calibration reported here is the model's own, not a wrapper's. Constraints are grounded in diabetes diagnostic guidance (American Diabetes Association 2025): soft constraints C_1 (glucose $\geq$ 126 mg/dL) and C_2 (BMI $\geq$ 30), and hard constraint C_3, a negative prediction under glucose $\geq$ 180 mg/dL. Pre-declared thresholds: ECE $<$ 0.10, abstention $\tau = 0.65$, satisfaction-score threshold 0.5 for the differentiable arm.

Provenance of this appendix. The original Pima harness was never committed, so the code released with this paper (`pima_pipeline/scripts/10-12`) is a reconstruction from the specification above rather than a recovered original. Two reported quantities depend only on the split and not on the model — the soft-constraint firing counts — which lets the original split be identified rather than guessed: sweeping seeds 0–399 across stratification and imputation-scope choices, exactly one configuration reproduces both counts *and* the reported headline triple. Every number below is the output of that released code.

A caveat on the constraints themselves. The Pima `Glucose` column is a two-hour oral-glucose-tolerance-test value, whereas the 126 and 180 mg/dL thresholds come from *fasting* glucose guidance. The rules therefore illustrate the protocol's mechanics on a public benchmark; they are not valid diabetes criteria, and we do not present them as such.

On the reference split the model attains AUROC 0.836 and accuracy 0.745 and passes all three axes: replay is bit-reproducible with a matching parameter hash (T1); there are zero hard violations, with the soft constraints firing on 94/231 and 137/231 of decisions (T2); and ECE is 0.070 with shift-sensitive abstention (T3).

The T3 pass is borderline, and we report it as such. Across 20 random splits T1 is stable (zero mismatches in all 20), but the bootstrap 95% CI on ECE ($B = 2000$ on the reference split) is $[0.041, 0.123]$, whose upper end exceeds the pre-declared 0.10 bound. The point estimate passes; the interval shows the pass is not robust to resampling, and a deployer should treat a borderline T3 as a prompt for recalibration rather than a certificate.

Quantity	Mean (20 seeds)	Interval
AUROC	0.839	$[0.802, 0.861]$
Accuracy	0.764	$[0.740, 0.788]$
ECE (multi-seed)	0.048	$[0.026, 0.061]$
ECE (bootstrap)	0.070	$[0.041, 0.123]$
T1 mismatches	0.0	$[0, 0]$
Hard violations	0.05	$[0, 1]^{\dagger}$
Abstain (shift)	0.306	$[0.255, 0.343]$
Abstain (normal)	0.186	$[0.151, 0.222]$

Multi-seed intervals are 5th–95th percentiles across seeds; the bootstrap row is a 95% percentile interval. †Range over seeds: a single violation in one seed, none in the other 19.

Table S5: Statistical robustness on the Pima task. Multi-seed means are over 20 stratified splits (seeds 0–19); the bootstrap interval is $B = 2000$ on the reference split, which is evaluated separately and is not folded into the sweep. The ECE interval crosses the pre-declared 0.10 bound, so the T3 pass is borderline and reported as such.

Negative controls. An uncalibrated control (probabilities sharpened upstream of the metric, so the deficiency is not tautological with the test) raises ECE from 0.070 to 0.199 and T3 flags it. A naturally deficient control — retrained with the glucose feature ablated, so it cannot represent the quantity the hard constraint depends on — drops to AUROC 0.753 and commits 8 hard violations on severe-hyperglycaemia cases, and T2 flags it. We prefer this to a forced-violation control, which would detect its own construction.

The concept-semantics probe separates the two cleanly: sweeping glucose from 70 to 200 mg/dL while holding the other features fixed, mean predicted risk rises monotonically from 0.074 to 0.846 in the genuine model (sensitivity 0.772) against a flat 0.338 in the glucose-blind one (sensi-

Deficient model	Effect	Suite
NC-A uncalibrated	ECE $0.070 \rightarrow 0.199$	T3 FLAGS
NC-B glucose-blind	AUROC $0.836 \rightarrow 0.753$; 8 violations	T2 FLAGS

Table S6: Negative controls (Pima, reference split). Both deficient models are flagged. NC-A injects miscalibration upstream of the metric; NC-B is a naturally deficient model, avoiding the circularity of a forced-violation control.

tivity 0.000 by construction, since the feature is absent). The glucose-blind model therefore satisfies or violates the constraint by accident rather than by dependence on the constrained concept.

Two integration styles. Attaching the same clinical implication as a fuzzy first-order axiom trained jointly with the network, rather than as a discrete post-hoc check, changes what T2 can report. The differentiable arm reaches comparable accuracy but emits a *continuous* satisfaction score, so converting T2 into a categorical verdict requires an additional thresholding decision that the pipeline-coupled arm never has to make. This is the differentiability–verifiability tension in miniature: soft constraints are convenient to optimise but do not natively yield the discrete, replayable checks categorical clinical auditing expects.

	Pipeline	Differentiable
AUROC / Acc.	0.836 / 0.745	0.817 / 0.745
T1 mismatches	0/231	0/231
T1 hash match	yes	yes
T2 constraint	discrete: 0 viol.	soft: 0.806 sat.
T3 ECE	0.070	0.111

T1 is two checks, both run for both arms: inference replay on the fitted model, and an independent refit from the same seed whose parameter hash must match. The differentiable arm's ECE exceeds the 0.10 bound, so on this run it does *not* pass T3 — the discrete arm does. Satisfaction is the mean over the 16 severe-hyperglycaemia test cases; at the 0.5 threshold it flags 3 of them.

Table S7: Two integration styles on the Pima reference split. The differentiable model yields a soft constraint-satisfaction score rather than a discrete check. Unlike the rest of this appendix, the differentiable arm is a *fresh run*, not a reproduction: no hyper-parameters for the original were recorded, so it uses the released implementation's defaults.

S3 Integration Styles and the Differentiability Trade-off

The differentiable arm is quantified in Table S7: a Logic Tensor Network (Badreddine et al. 2022) in which the same constraint (severe hyperglycaemia implies elevated risk) is a fuzzy first-order axiom trained jointly with a neural predicate via a satisfaction loss. Two consequences are worth stating beyond the table.

First, the axis that separates the two interfaces is T2, not accuracy. The pipeline-coupled model emits *discrete* per-decision checks, so its T2 result is a count that either is or is not zero. The differentiable model emits a *soft* satisfaction score, so a categorical verdict requires an extra thresholding decision that is itself a modelling choice — and the threshold is not implied by the guideline the constraint encodes. The protocol accommodates both interfaces, but the appropriate T2 criterion differs by interface type, and standardising satisfaction-score thresholds for differentiable systems remains open.

Second, on this run the differentiable arm does not pass T3 (ECE 0.111 against the 0.10 bound) while the discrete arm does. We do not read this as evidence that differentiable integration calibrates worse in general — it is one run of one architecture on one split, and the arm is a fresh run rather than a reproduction of the original, whose hyper-parameters were never recorded. We report it because suppressing an axis failure in the appendix while reporting the same axis in the body is exactly the selective reporting this framework exists to prevent.

S4 A Representative Decision Trace

The trace below is *synthetic*: it is structurally identical to a real trace but contains no patient data, since the PhysioNet DUA bars release of per-decision derivatives. Each decision carries a decision identifier, a model-version hash, a configuration hash, the sub-symbolic output, the prediction, and the per-constraint boolean evaluations — the artifacts T1 and T2 consume. The symbolic structure is not implicit in the weights; it is declared, version-controlled and logged, which is what makes the decision auditable.


```
{"decision_id": "D0000",
 "model_version": "a8a90b255783d901",
 "config_hash": "2e017e520afd7554",
 "neural_proba": 0.240617,
 "prediction": 0,
 "constraint_checks": {
   "C1_glucose_threshold": true,
   "C2_bmi_obese": false,
   "C3_violation_neg_high_glucose": false}}
```


S5 Threshold and Severity Sensitivity

Certificate thresholds. Sweeping the coverage floor over $\beta \in [0.30, 0.70]$ and the deployed-violation cap over $\{0.02, 0.05, 0.10\}$, no policy passes at any of the 30 grid points. The logged policy would require a cap above 0.74 and the enforced policy a floor below 0.20; neither is clinically defensible.

Severity definition. Table S9 reports the pre-declared sensitivity arm over 20 seeds. The primary v1.3 specification (absolute limb only) remains primary; adding the relative limb ($\geq 3\times$ baseline) moves neither the ICD agreement nor the violation rate materially, and the certificate fails under all three definitions. At cohort level the primary defi-

Policy	Coverage	Viol. rate	Certificate
Logged	0.78	0.74	FAIL (violation)
Enforced	0.20	0.00	FAIL (coverage)
Floor / cap	≥ 0.50	≤ 0.05	—

Table S8: Compliance–coverage certificate on the acute-severe subgroup (20 seeds, conservative across-seed bounds: coverage is the 5th percentile and the violation rate the 95th; the corresponding means are 0.78/0.71 and 0.23/0.00). Neither policy passes: the logged policy violates, the enforced policy under-covers.

nition yields 5,954 severe cases at agreement 0.3697, the extended definition 6,091 at 0.3722, and the relative limb alone 211 at 0.4645; the 137 cases newly captured by the relative limb have agreement 0.4818. These are *cohort-level* counts over all 378,591 admissions; Table S9 reports the same three definitions as *per-split* means over the 20 test folds, so its figures differ (63.8 relative-limb cases at agreement 0.4633 rather than 211 at 0.4645) — the same quantity on a different denominator, not a disagreement. The relative limb uses the within-24h nadir as a proxy for pre-morbid baseline, inheriting the same conservative bias the primary arm discloses. RRT initiation is not computable (it requires MIMIC-IV procedure tables outside the cohort derivative we build) and the paediatric eGFR limb does not apply to an adult-only database; both omissions make the reported gap conservative. These rules are the authors' reading of published KDIGO criteria and are not clinician-ratified.

Definition	n severe	ICD agr.	Viol. rate
Absolute (primary)	1772.8	0.3759	0.7143
Absolute or relative	1815.2	0.3783	0.7160
Relative only	63.8	0.4633	0.7504

Table S9: Severity-definition sensitivity, 20 seeds, per-split means. The primary specification is unchanged; the certificate fails under all three definitions.

S6 Extended Results

Failure-mode taxonomy, full cells. Table S10 gives the marginals of the two-axis partition of confident denials over 20 seeds (total severe 1772.85, total confident denials 1011.95 per split). The joint cells are in the released `error_taxonomy.json`.

Label validity by subgroup, full table. Table S11 gives the model-free agreement rates with patient-level cluster bootstrap intervals ($B = 2000$) on all 5,954 severe cases. Two race buckets are reported for completeness but rest on small counts: AIAN/NHPI ($n = 42$, agreement 0.1905, CI 0.0769–0.3429) and Unknown/Declined ($n = 366$, 0.5519, 0.5000–0.6067). The latter is the highest agreement of any bucket, which we read as a data-recording artifact rather than a clinical signal and do not interpret.

Model-dependent subgroup quantities. Per the regis-

Category	Denials	Severe	Rate
By label status			
No AKI code	888.1	1107.1	0.802
AKI code present	123.9	665.8	0.186
By creatinine trajectory			
Elevated at arrival	870.5	1344.2	0.647
Rise, elevated base	136.3	408.5	0.334
Rise, near-normal base	5.2	20.2	0.255

Table S10: Confident no-AKI decisions on guideline-severe cases, mean per split over 20 seeds. Rate is the mean of the within-category rate across seeds, so it can differ in the third decimal from the ratio of the two mean counts.

Subgroup	n	Agreement	95% CI
All severe cases	5954	0.3697	0.3533–0.3855
White	2978	0.4396	0.4150–0.4625
Black	1677	0.2767	0.2497–0.3049
Hispanic/Latino	400	0.2050	0.1557–0.2634
Asian	242	0.2438	0.1938–0.3022
Other/Multiple	249	0.3092	0.2471–0.3736
Male	3857	0.3863	0.3665–0.4070
Female	2097	0.3391	0.3153–0.3656
Age 18–44	1145	0.3031	0.2700–0.3423
Age 45–59	1870	0.3540	0.3266–0.3836
Age 60–74	2035	0.3799	0.3543–0.4070
Age 75+	904	0.4635	0.4253–0.5011
Medicare	3593	0.3198	0.3010–0.3400
Private	1259	0.4647	0.4318–0.4963
Medicaid	874	0.4245	0.3834–0.4669
Other insurance	179	0.3408	0.2640–0.4186

Table S11: Fraction of guideline-severe cases carrying an ICD AKI code, by subgroup, on all 5,954 cases (3,928 patients). Model-free: no training, split or seed. Intervals are patient-level cluster bootstrap, $B = 2000$.

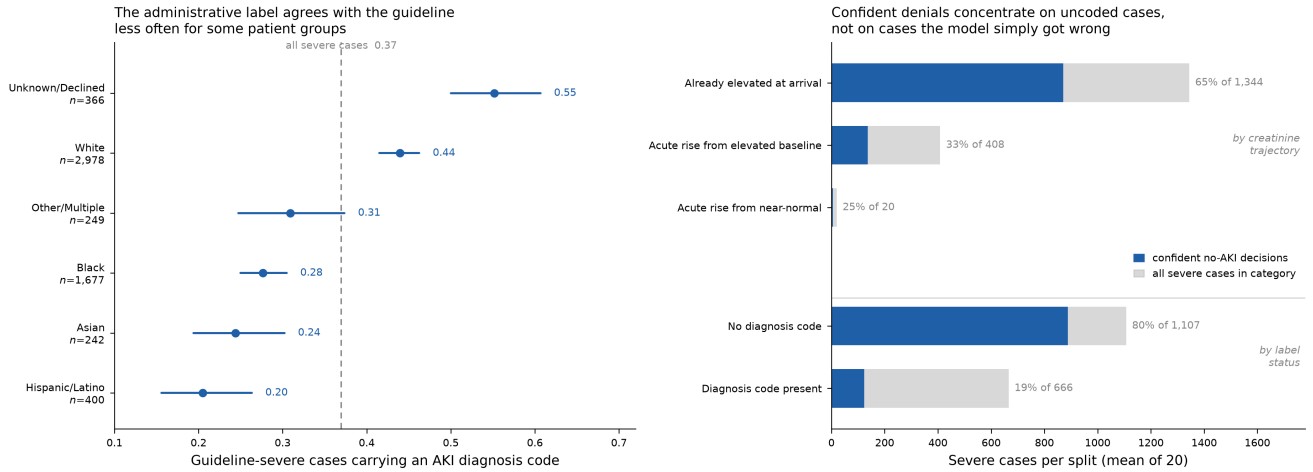


Figure S1: Left: fraction of guideline-severe cases carrying an ICD AKI code, by race bucket, on all 5,954 severe cases with patient-level cluster bootstrap 95% intervals ($B = 2000$); the dashed line is the cohort-wide 0.37. Right: confident no-AKI decisions among severe cases, mean per split over 20 seeds, split by administrative label status and by creatinine trajectory. Denials concentrate on cases the label missed, not on cases the model simply got wrong. (Reproduced as Figure 2 of the main paper; kept here so that the S-numbering of later figures is unchanged.)

tered floor of $n \geq 500$ decisions per split, only five subgroups are determinate: age 60–74 (mean $n = 608.5$ per split; deployed violation rate 0.6919, 5th–95th percentile 0.6479–0.7426), Male ($n = 1148.4$; 0.7012, 0.6569–0.7344), Female ($n = 624.5$; 0.7375, 0.7040–0.7656), White ($n = 884.8$; 0.6287, 0.5943–0.6635) and Medicare ($n = 1073.1$; 0.7425, 0.7101–0.7809). All others, including Black patients ($n = 497.1$), are reported UNDETERMINED and no comparison is drawn from them.

Arm	AUROC	Viol.	Cov.	Cert.
MLP (audited)	0.8797	1012	0.78	FAIL
GBM	0.8815	1028	0.79	FAIL
Logistic	0.8062	753	0.66	FAIL
Pure-neural	0.8797	n/e	n/c	n/c

n/e: no constraint evidence emitted. n/c: not computable.

Table S12: Model arms on identical 20 patient-level splits of MIMIC-IV. Accuracy separates the arms; compliance does not. **Viol.** is the mean per-split hard-violation count and **Cov.** the 5th-percentile deployed coverage on the severe subgroup; the certificate additionally requires a 95th-percentile deployed violation rate ≤ 0.05. The pure-neural arm wraps the same predictor and emits no constraint evidence, so its compliance is not computable — which is not the same as passing.

Chronicity proxy: threshold sensitivity and model specification. Table S14 reports the proxy analysis of Section 5.3 at the primary cut and three alternatives. The primary cut is 1.15; the alternatives were computed in the same run and are reported so the verdict is not resting on one threshold, but they are a sensitivity check and not eight independent tests. At every cut the flat-high stratum consists exclusively of cases qualifying through the absolute acute-rise limb.

The logistic robustness fit quoted in Section 5.3 is specified as follows, so it is reproducible from the released code (`08_stratified_label_validity.py`, block F). *Outcome:* the ICD AKI label itself, 1 if the admission carries an ICD-9 584.x or ICD-10 N17.x code. *Sample:* the 4,655 guideline-severe admissions in the White and Black buckets; the remaining 1,299 severe cases fall in four smaller buckets (Hispanic/Latino 400, Unknown/Declined 366, Other/Multiple 249, Asian 242, AIAN/NHPI 42) and are excluded from this two-group contrast rather than pooled. *Covariates:* a Black indicator, two trajectory-band indicators (A3, A2, with A1 as reference), the chronicity-proxy indicator, and a race-by-proxy product term. *Missingness:* none — the outcome, both creatinine values and the race field are complete on all 4,655 rows, so no imputation is applied; the proxy ratio is defined as 1.0 where the nadir is zero, which occurs in no severe case. *Inference:* the same patient-level cluster bootstrap used elsewhere in this paper ($B = 400$, resampling `subject_id`), refitting the model on each draw, rather than a model-based standard error, so the clustering assumption does not change between analyses. Coefficients: Black -0.66 (odds ratio 0.52, CI -0.89 to -0.45); A3 $+0.91$ (2.48, 0.76–1.08); A2 $+0.70$ (2.02, 0.11–1.25); chronic-like $+0.41$ (1.50, 0.24–0.57); race$\times$proxy $+0.04$ (1.04, -0.23 to 0.28). This is a conditional association, not a causal adjustment.

Per-axis verdicts and audit-readiness. Table S15 reports the three axes separately for both deployment policies, with the readiness count never collapsed across them. The enforced policy reaches 3/3 only because escalation makes T2 pass by construction, at the cost of deferring 57% of acute-

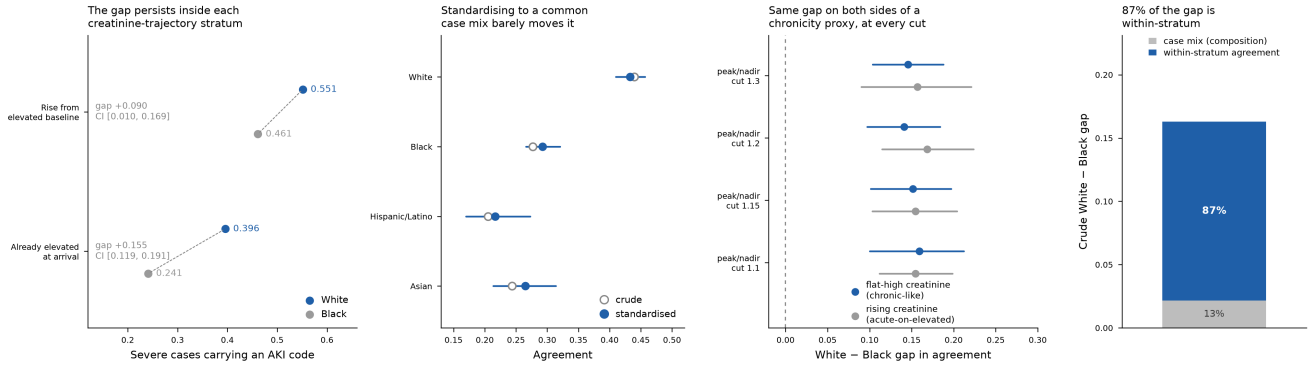


Figure S2: Does measured case mix explain the racial gap in label validity? All intervals are patient-level cluster bootstrap, $B = 2000$. (1) Agreement within each creatinine-trajectory stratum: the gap persists inside both adequately sized strata. (2) Crude agreement (open) against agreement standardised to the cohort's trajectory distribution (filled). (3) The White−Black gap in each stratum of a within-window chronicity proxy (24h peak/nadir), at the primary cut and three alternatives: every interval excludes zero and the strata are similar, with no evidence of interaction (no equivalence test was performed). The proxy is *not* a CKD measurement — a same-admission CKD code is assigned post-discharge by the process under audit, and prior-admission codes were not available to this analysis; the prior-admission adjustment is in Appendix S7. (4) Kitagawa decomposition of the crude difference into composition and within-stratum terms. All exploratory, not pre-registered.

Stratum	**Group**	n	**Agreement**	**Difference (95% CI)**
Elevated at arrival	White	2128	0.396	
	Black	1395	0.241	+0.155 (0.120–0.191)
Rise, elevated base	White	811	0.551	
	Black	269	0.461	+0.090 (0.010–0.169)
Rise, near-normal	White	39	0.513	
	Black	13	0.308	below floor, not tested
Standardised to the cohort trajectory mix				
White		2978	0.440 → 0.433	(0.410–0.457)
Black		1677	0.277 → 0.292	(0.266–0.321)
Hispanic/Latino		400	0.205 → 0.216	(0.170–0.273)
Asian		242	0.244 → 0.265	(0.214–0.314)

Table S13: Stratified and standardised label validity behind Figure S2. Upper: agreement within each creatinine-trajectory stratum. The A2 row falls below the pre-registered subgroup floor and is reported without a tested gap. Lower: crude against standardised agreement, standardising each group to the whole cohort's trajectory distribution. Intervals are patient-level cluster bootstrap ($B = 2000$). Exploratory.

Cut	**Stratum**	n	**White**	**Black**	**Difference (95% CI)**
1.10	flat-high	1649	0.431	0.272	+0.159 (0.100–0.212)
1.10	rising	2862	0.376	0.221	+0.155 (0.112–0.199)
1.15	flat-high	2558	0.425	0.274	**+0.152** (0.101–0.197)
1.15	rising	1953	0.354	0.199	**+0.155** (0.103–0.204)
1.20	flat-high	3072	0.417	0.276	+0.141 (0.097–0.184)
1.20	rising	1439	0.343	0.175	+0.169 (0.115–0.224)
1.30	flat-high	3615	0.412	0.266	+0.146 (0.104–0.188)
1.30	rising	896	0.316	0.160	+0.157 (0.090–0.221)

Table S14: White−Black difference in agreement within each chronicity-proxy stratum, among the 4,511 elevated-at-arrival severe cases. Intervals are patient-level cluster bootstrap ($B = 2000$). The primary cut is 1.15; the other three rows are a sensitivity check. Similarity across strata is descriptive — no equivalence test was performed.

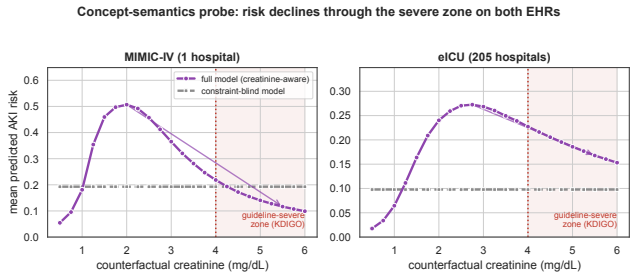


Figure S3: Concept-semantics probe on both databases. Mean predicted AKI risk under a counterfactual creatinine sweep rises then *declines* through the guideline-severe zone on MIMIC-IV (1 hospital) and eICU (205 hospitals): the model uses creatinine in the clinically wrong direction at the severe end, tracking each database's under-coding of high-creatinine AKI.

severe cases; the certificate exists precisely so that this route does not read as a clean pass.

Axis / quantity	Logged	Enforced
AUROC	0.8797 [0.8779, 0.8817]	
T1 reproducibility	PASS (20/20)	
T3 ECE (95% CI upper)	0.0043 — PASS	
T2 hard violations (mean)	1012	0
T2 result	FAIL (0/20)	PASS
Escalation on severe	—	57%
Audit-readiness	**2/3**	**3/3**

Table S15: Per-axis verdicts on MIMIC-IV (20 patient-level splits). The AUROC bracket is the range over seeds, not a confidence interval. The readiness count is reported per policy and never collapsed into one number.

Per-axis stability. Across the 20 patient-level splits the logged deployed-violation rate (0.71) and enforced coverage (0.23) vary little and sit far from any passing threshold, so the verdicts are not split-dependent. The abstention threshold τ takes 19 distinct values across the 20 seeds over the range 0.6441–0.6587 (see Appendix S1, row 7). Figure S4 shows the per-axis distributions behind that statement.

Calibration behaviour. Figure S5 shows that the calibration axis discriminates calibrated from miscalibrated behaviour rather than passing trivially: the deployed isotonic-calibrated model tracks the diagonal, while the over-confident negative control NC-A departs from it and is flagged.

The certificate, geometrically. Figure S6 plots the admissible region of Table S8: a policy passes only inside it, and both measured operating points fall outside on opposite sides, which is the property that makes the verdict robust to gaming by deferral.

S7 Prior-CKD Sensitivity: the Analysis We Said We Could Not Run

An earlier revision named the missing prior-CKD adjustment as a limitation and released `10_prior_ckd_sensitivity.py` unrun, because the two credentialed tables it needs (`diagnoses_icd`, `admissions`) were not available where that revision was produced. They since became available. The analysis is **exploratory, post hoc and selection-limited**, and its two findings do not have the same status, so we separate them.

Finding 1 comes from the script exactly as released: blocks A–E were not touched, and a line-level diff against the released version shows 97 lines added and none removed or altered. *Finding 2 does not.* Running block E showed that most of the cohort carries a prior renal-replacement code, which we had not anticipated; we then *added* block G to measure what that implied, and after seeing its first output we rewrote its interpretation, because our initial reading — that the rule fires on noise in dialysis-dependent patients — was contradicted by the rise profile the same block reports. Finding 2 is therefore post-hoc analysis chosen after seeing results, one step further from pre-specification than Finding 1, and we mark it as such wherever it appears. No registered threshold was altered by either (Appendix S1, row 15).

Design. The CKD indicator is built *only* from admissions that discharged before the audited admission started. A same-admission CKD code is never used: MIMIC-IV diagnosis rows are assigned after discharge from the signed record, so conditioning one product of the coding process on another would not estimate what it appears to — a hazard documented at scale, with 37 of 92 MIMIC same-admission prediction studies using such codes as features (Ramadan et al. 2025). The indicator is named for what it measures, *CKD documented in a prior MIMIC hospitalisation*, not *has CKD*: care outside the MIMIC window is invisible.

Provenance. The raw tables reproduce our cohort exactly: all 378,591 cohort admissions are present, `admittime` matches on 100% of them, and recomputing the AKI label from `diagnoses_icd` agrees with the built cohort on 100.000% of admissions at the same 0.1834 base rate.

Selection. Of 5,954 guideline-severe cases, 4,203 (70.6%) have an earlier MIMIC admission and enter the analysis. That restriction is *unequal*: 82.9% of severe Black against 69.7% of severe White cases have observable history, a 13.2-point spread, in the direction of the gap under study. Results below therefore describe patients with prior MIMIC contact, not all severe cases, and are a selection-limited sensitivity analysis rather than a correction of the main estimate. (An earlier revision quoted 80.9% and 67.9% for the same quantity, from a proxy in `08_stratified_label_validity.py` that ordered admissions *within the cohort*. This script uses the full `admissions` table and requires the prior discharge to precede the current admission, which finds slightly more history; the figures above are the ones the analysis actually ran on.) Among those with history, prior CKD is documented for 92.8% of Black and 84.9% of White severe cases.

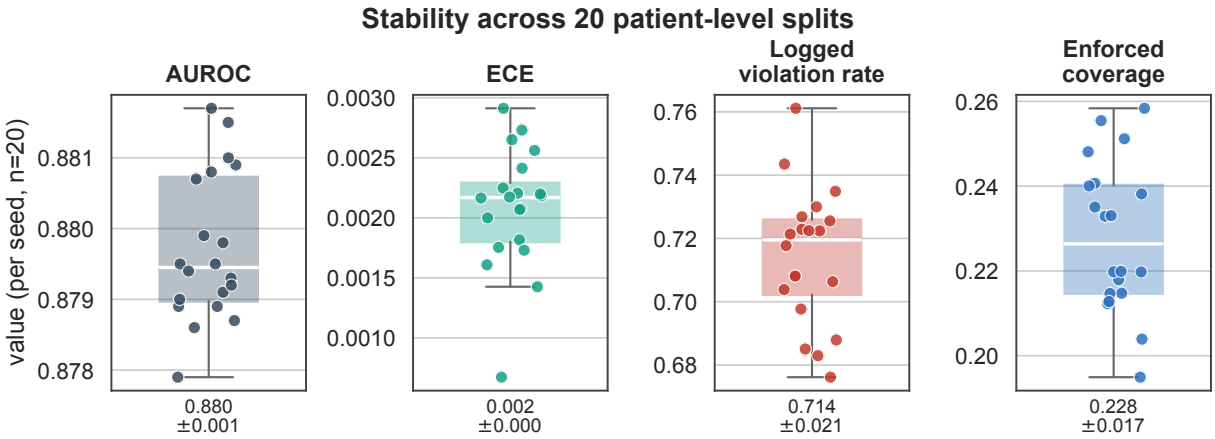


Figure S4: Per-axis stability across 20 patient-level splits on MIMIC-IV (each panel auto-scaled; mean±SD below). Discrimination and calibration are tight; the logged deployed-violation rate (0.71) and enforced coverage (0.23) are stable and far from any passing threshold, so the per-axis verdicts are not split-dependent.

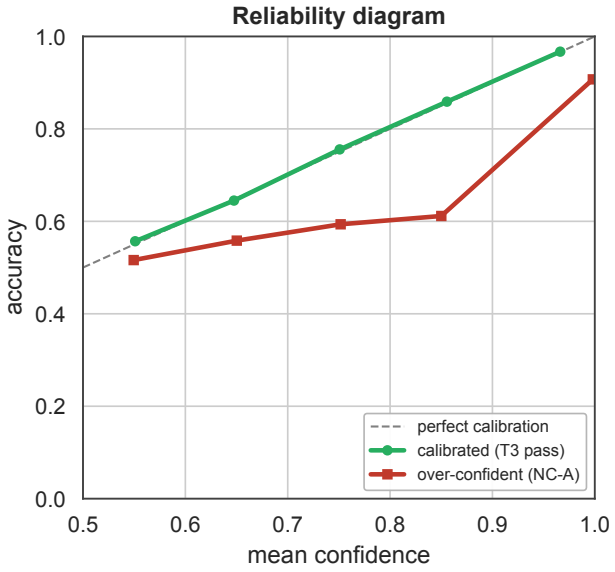


Figure S5: Reliability diagram (MIMIC-IV). The deployed isotonic-calibrated model tracks the diagonal (T3 pass); the over-confident negative control NC-A departs from it and is flagged by T3.

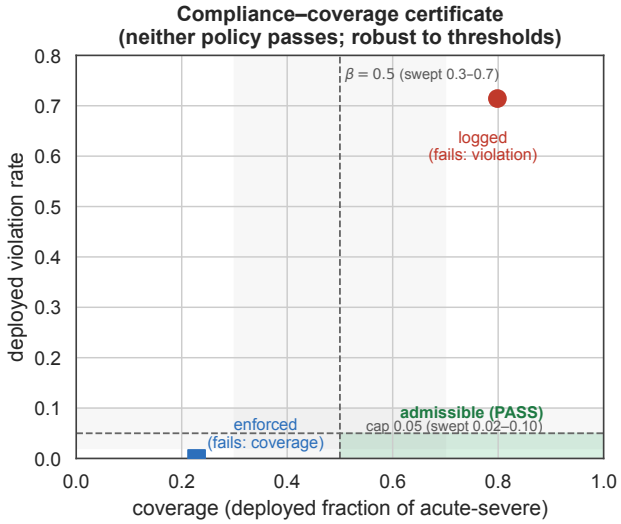


Figure S6: Compliance–coverage certificate on the acute-severe subgroup. A policy passes only inside the admissible region (coverage $\geq \beta = 0.5$ *and* deployed violation rate ≤ 0.05). Both measured operating points fall outside: the logged policy violates (high, right), the enforced policy under-covers (low-left); shaded bands show the swept threshold ranges over which the verdict is unchanged.

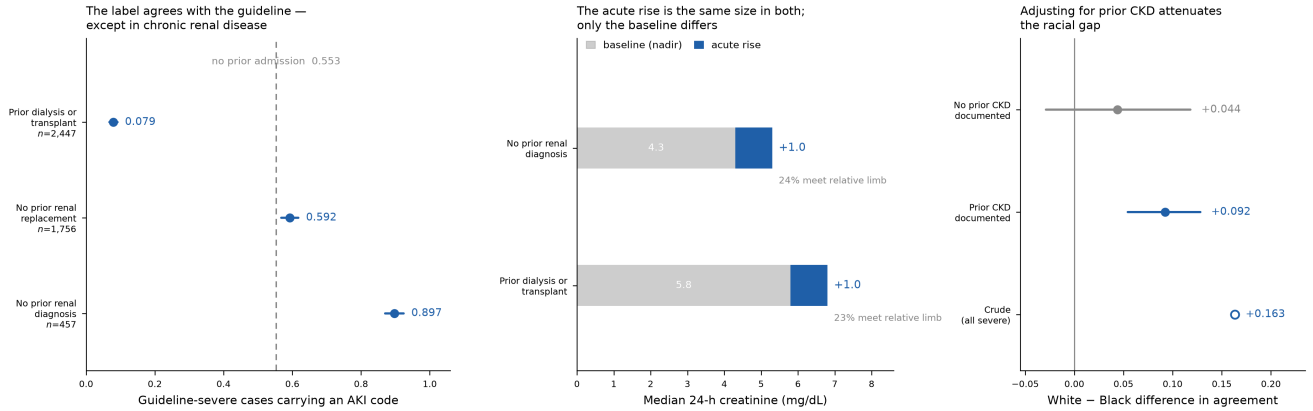


Figure S7: Prior-CKD sensitivity (Appendix S7), exploratory and selection-limited to the 4,203 of 5,954 guideline-severe cases with an earlier MIMIC admission. All intervals are patient-level cluster bootstrap, $B = 2000$. Left: the label–guideline disagreement is concentrated in chronic renal disease. The three plotted strata are nested, not disjoint: "no prior renal diagnosis" ($n = 457$) is a subset of "no prior renal replacement" ($n = 1{,}756$), and together with the prior-replacement stratum ($n = 2{,}447$) these cover the 4,203 cases with observable history. The dashed line marks the 1,751 severe cases with no prior admission, whose renal history is unknowable. Centre: the acute rise is the same median size in both extremes and nearly the same share meets the guideline's relative limb, so the rule is not firing on noise — only the baseline differs. Right: the White−Black difference in agreement, crude and within prior-CKD strata; filled markers exclude zero, open marker is the unadjusted comparison, and grey does not exclude zero.

Finding 1: the racial gradient survives adjustment, but shrinks. (Figure S7, right.) Within the prior-CKD stratum the White−Black difference in agreement is $+0.092$ (95% CI 0.054–0.128, $n = 3{,}736$), against a crude $+0.163$ in the full severe population. In the much smaller no-prior-CKD stratum it is $+0.044$ (-0.029–0.118, $n = 467$) and no longer excludes zero. Double stratification by creatinine trajectory agrees: $+0.073$ (0.036–0.111, $n = 2{,}976$) among elevated-at-arrival cases with prior CKD and $+0.124$ (0.031–0.210, $n = 737$) among those rising from an elevated baseline, with both no-prior-CKD cells indistinguishable from zero. Standardising to the joint (trajectory × prior-CKD) mix moves White from 0.372 to 0.341 and Black from 0.232 to 0.263 — a real narrowing the trajectory-only standardisation in Section 5.3 did not show. In a logistic fit carrying trajectory, the chronicity proxy and prior CKD together, the Black indicator is no longer distinguishable from zero (odds ratio 0.591, log-odds CI -1.197 to 0.301, $n = 3{,}465$, the White and Black cases among the 4,203 with observable history), and the race-by-CKD interaction is null (1.085, -0.794 to 0.758). **We therefore weaken the subgroup claim.** Documented pre-existing CKD explains part but not all of the gradient: the stratified differences stay positive and mostly exclude zero, the adjusted regression does not, and the gradient is not established as independent of prior CKD.

Finding 2, unanticipated and added after seeing results: the headline gap is concentrated in chronic renal disease, and that limits our constraint as much as the label. (Figure S7, left and centre; block G, added post hoc.) Agreement among severe cases with *no* prior renal diagnosis of any kind is **0.897** (CI 0.870–0.924, $n = 457$); among those with a prior dialysis or transplant status code it is **0.079** (0.068–0.091, $n = 2{,}447$), intervals non-overlapping. The obvious reading — that our rule fires on noise in dialysis-dependent patients — is *wrong*, and our own data say so: the median absolute rise is 1.0 mg/dL in both strata and nearly the same share meets the relative limb (23.2% against 23.6%). What differs is the baseline the rise sits on (median nadir 5.8 against 4.3 mg/dL). The rule fires on a comparable acute change on top of a chronic baseline, and whether that should carry an AKI code is a coding-convention question we answer one way and coders answer the other. This was not anticipated by our pre-registration, it is a limit on the *constraint* as much as on the label, and it narrows the population to which our headline most confidently applies. A further 1,751 severe cases have no prior MIMIC admission at all, so their renal history is unknowable; their agreement is 0.553, between the two extremes, and they are reported so the three strata are not read as a partition of all severe cases.

Leave the stratum out entirely. The sharpest form of the objection is that the headline is an artifact of dialysis-dependent patients. That is testable by deletion rather than by argument. Three disjoint strata (prior renal replacement, $n = 2{,}447$; observable history without renal replacement, $n = 1{,}756$; no prior admission, $n = 1{,}751$) partition all 5,954 severe cases exactly, so the test is a reweighting of the same measurement. The prior-replacement stratum is **41.1%** of all severe cases — the 58% figure that can be read off block G uses the observed-history denominator, not the full severe cohort. Deleting it leaves **3,507** severe cases, 59% of the total, at agreement **0.5726**: the administrative label still disagrees with the guideline on **43%** of them. The concentration finding narrows which population the headline describes best; it does not make the headline an artifact of that population. (Block H, computed in the same script; it is a

deterministic reweighting of block G and introduces no new model or threshold.) The same deletion applied to the *model* statistic — the post hoc restricted-cohort certificate, with patient-clustered intervals — is in Section S14, Tables S17 and S18: the deployed violation rate falls from 0.71 to 0.53 and the certificate still fails on the violation limb.

What does not change. The compliance failure is a property of the decisions the model deploys, not of which patients are in the cohort: the certificate fails under every severity definition tested (Appendix S5), and on the matched positive control the failure disappears when only the training target changes (Section 5.5). Finding 2 sharpens where the label–guideline disagreement lives; it does not restore agreement between an ICD-trained model and a guideline rule.

S8 The eICU Replication Arm

The body reports eICU in one paragraph; this appendix states what was built and what was measured, so the replication can be read on its own terms (`eicu_pipeline/`; aggregate results in `results/audit_summary_eicu_v1.3.json`).

Cohort and label. The unit of analysis is the ICU unit stay (`patientunitstayid`): 169,825 stays grouped into 147,832 hospital stays (`patienthealthsystemstayid`, used as the split group) across 205 hospitals. The label is a diagnosis-string proxy for the administrative AKI diagnosis: a stay is positive if any entry in the eICU `diagnosis` table matches *acute renal failure* or *acute kidney* (case-insensitive), and negative otherwise. It is derived from the diagnosis table only, never from creatinine, so it is independent of the constraint in the same sense as the MIMIC-IV ICD label. The base rate is 10.9% against 18.3% on MIMIC-IV; eICU diagnosis strings are not ICD codes and are entered by clinical staff rather than coders, which is what makes this a second administrative labelling system rather than a second copy of the first.

Features and constraint. The feature set mirrors MIMIC-IV exactly: the first value within the first 24 hours after unit admission for the same eight chemistry analytes (creatinine, glucose, BUN, potassium, sodium, haemoglobin, white cell count, bicarbonate), plus age and sex, with missing values median-imputed. The acute-severe mask is the same rule at the same thresholds (first-24h peak creatinine ≥ 4.0 mg/dL with a rise $\geq 1.5\times$ nadir or ≥ 0.3 mg/dL), giving 4,142 acute-severe stays, of which 43.2% carry the proxy label. The model, calibration wrapper, harness, pass thresholds and certificate constants are the frozen `prereg-v1.3` configuration; only the cohort adapter is dataset-specific.

Results. Over the same 20-seed protocol: AUROC 0.852 (range 0.849–0.856), T1 20/20, T3 ECE CI upper 0.006 against the 0.10 bound. On the acute-severe subgroup the model deploys far less than on MIMIC-IV: logged coverage is 0.39 (5th percentile 0.34), and 0.93 of those deployed decisions are confident denials (95th percentile 0.98), a mean of 442 hard violations per split. The logged policy therefore fails the certificate on both limbs, the enforced policy escalates 36% of severe stays on top of the existing abstention and covers 0.03 (5th percentile 0.01), and neither passes. The higher abstention reflects the lower base rate: calibrated probabilities on a 10.9%-prevalence task sit closer to the abstention band, so more severe stays fall below τ. This is a difference in the failure surface, not in the verdict, and we report it rather than fold it into the MIMIC-IV numbers.

Shift axis. The distribution-shift evaluation halves the 205 hospitals at random (seeded), trains on one half and evaluates on the held-out half. On this genuinely institutional shift the deployed model keeps calibration (held-out-hospital ECE CI upper 0.005 to 0.009 across three seeds) and abstains slightly more on unseen hospitals (direction sub-criterion 3/3), whereas the MIMIC-IV era shift gives 0/3 on direction. Neither result enters a verdict, because the T3-shift sub-criterion is unvalidated by negative control (Appendix S9, item 5). The severe-subgroup certificate on held-out hospitals, which this shift arm did not report at submission, is now in Section S14, Table S19: on 103 held-out hospitals discrimination transfers (AUROC 0.849) and the certificate fails on coverage, with every per-site verdict UNDETERMINED under the registered floor.

Controls and limits. NC-B (constraint-blind), NC-C and NC-D behave as registered; NC-A does not fire at the registered severity (item 4 of Appendix S9; Table S2 row 8). Two limits are specific to this arm: splits group by hospital stay rather than by patient, so a readmitted patient can straddle a split, and the proxy label's string match was verified against the eICU diagnosis vocabulary but has no external validation study analogous to those cited for ICD codes.

S9 Measurements Not Reported in the Body

These four measurements were made under the registered protocol and are reported here rather than in the body, where the page limit did not allow them. Three bear on how conservative the paper's headline figures are; the fourth is a methodological finding about negative controls that we think generalises beyond this study. None of them was omitted because it was unflattering, and one of them (§S9.4) is unflattering.

1. The creatinine-only definition under-counts AKI, measured. Our severity rule implements the serum-creatinine limb of KDIGO only. Because KDIGO defines AKI by *either* limb, that omission can only under-count, and the body says so; here we measure by how much. The urine limb is computable for the 51,089 admissions with an ICU stay, where hourly output is charted. Under a deliberately conservative rule — full 24-hour charting and at least two low 6-hour blocks, i.e. $\geq$12 hours of sustained oliguria at < 0.5 mL/kg/h — the urine criterion flags **6,253 additional AKI cases with no qualifying creatinine rise**, against 16,899 creatinine-positive admissions in the same subset: a **37%** under-count on the lower bound. A permissive rule (any single low 6-hour block) gives 16,399 additional cases, a 97% under-count, which we regard as an upper bound inflated by sparse charting and transient dips. Both bounds point the same way for the *AKI-positive population*: the guideline-positive set we audit against is *smaller* than the true one. Whether the omitted cases would raise or lower the severe-subgroup label-

validity gap and violation rate is unevaluated, and the body (Section 6) therefore no longer describes the omission as conservative (the earlier wording is retained in Section S13). Three caveats: the ICU window is anchored on ICU admission while the creatinine label is anchored on hospital admission, the blocks are fixed rather than rolling, and the limb is computable only for admissions with an ICU stay (`results/urine_output_arm.json`).

2. The certificate verdict does not depend on where we set the thresholds. The compliance–coverage certificate takes two pre-registered constants: a coverage floor β and a deployed-violation cap. A reader may reasonably ask whether the failure is an artifact of those choices. We swept $\beta \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ against caps $\{0.02, 0.05, 0.10\}$ for both deployment policies — 30 grid points. **No policy passes at any grid point.** Loosening the cap by a factor of five and the floor by twenty percentage points does not produce a passing configuration, because the logged policy's violation rate (0.74 at the 95th percentile) and the enforced policy's coverage (0.20 at the 5th; mean 0.23) are both far from any threshold on the grid rather than marginally the wrong side of one (`results/threshold_sensitivity_v1.3.json`).

3. An admission-level test of the subgroup gradient, and why we do not use it. A four-group χ^2 over the race buckets gives 187.9 on 3 degrees of freedom ($p \approx 2 \times 10^{-40}$). We report it for completeness and *not* as our test: it treats admissions as independent, and patients with repeated severe admissions contribute several correlated rows, which makes it anticonservative. The patient-level permutation test in Section 5.3, in which each patient contributes the mean over their own severe admissions, is the primary result and gives a far less spectacular p at the same effect size.

4. A fixed-severity negative control is not portable across datasets, and reporting the triggering severity fixes it. The uncalibrated control NC-A sharpens predicted probabilities by a fixed factor $k = 3$ and should breach the calibration bound. It does on MIMIC-IV (ECE CI upper 0.102 against the 0.10 bound) and *does not* on eICU (0.088) — the same control, the same code, opposite verdicts. This is not a property of the models: eICU's lower base rate (10.9% against 18.3%) concentrates predictions nearer the low end, where a multiplicative stretch moves them less. A pass/fail control calibrated on one dataset therefore silently becomes a non-control on another, which is a trap for anyone reusing a published audit suite. The repair is to report the *triggering severity* $k^\star$ — the smallest stretch whose ECE CI upper reaches the bound — instead of a fixed-k verdict. Measured on the same grid (k from 1.0 to 8.0 in steps of 0.25), $k^\star$ is **3.125** on MIMIC-IV (range 3.0–3.25 over 20 seeds) and **8.0** on eICU (7.25–8.0), and on 15 of 20 eICU seeds the bound is not reached anywhere on the grid, so the eICU figure is right-censored and the true separation is wider than $2.6\times$. $k^\star$ is a measurement that transfers; a fixed-k verdict is not. We did not pre-register $k^\star$; it was added after the discrepancy appeared, and the fixed-k control remains the registered one (`eicu_pipeline/results/ncA_kstar.json`).

5. Two registered negative controls did not fire, and on two different shift axes. The stale-model control NC-C was registered to trigger the shift sub-criterion of T3 and does not, on either database — but the two runs test *different* shifts, and the distinction matters for how much the null result means. On MIMIC-IV the axis is temporal: train on 2008–2013, evaluate on 2017–2022, leaving calibration essentially intact (late-window ECE CI upper 0.014, abstention 0.137). On eICU the axis is *institutional*: the 205 hospitals are randomly halved and the model is evaluated on the held-out half (`eicu_pipeline/run_eicu.py`), again leaving calibration intact (0.006, 0.077). The eICU run is therefore the stronger test of the two — a genuine multi-centre shift rather than an era shift within one institution — and it still does not fire, so we cannot attribute the null to a weak axis. For completeness, the audited model itself passes the abstention-direction sub-criterion on 3 of 3 eICU shift seeds but on 0 of 3 MIMIC-IV seeds (abstention falls from 0.16 to 0.14 across the era shift); since the sub-criterion is unvalidated, neither outcome enters a verdict. The T3-shift sub-criterion is recorded as **unvalidated by negative control** and excluded from every readiness count in this paper (Appendix S1, Table S3). One implementation note for anyone reading the code: the eICU harness encodes its hospital split in the column named `anchor_year_group`, reusing the temporal machinery rather than duplicating it, so that column name does *not* mean an era split there. NC-A's failure to fire on eICU is row 8 of Table S2. By the pre-registration's own §6 rule a control that does not fire counts against the framework, not for it.

S10 Positioning Relative to Prior Art

Table S16 states the delta against each line of prior work. The protocol reuses established statistical machinery and contributes no new estimator; what is new is the assembly into a pre-registered, per-decision, guideline-anchored clinical conformity test with negative-control validation and a coverage floor that resists gaming by deferral.

S11 Governance Mapping

Each axis produces evidence that a governance instrument calls for: replayable traces speak to EU AI Act Article 13 transparency (European Parliament and Council 2024) and NIST AI RMF documentation (National Institute of Standards and Technology 2023); per-decision constraint logs speak to FDA GMLP lifecycle and safety expectations (U.S. Food and Drug Administration 2025); calibration with abstention speaks to NIST AI RMF risk controls. Management-system standards such as ISO/IEC 42001 (International Organization for Standardization 2023) require documented, monitorable evidence of the same general kind.

We state the limits of this mapping explicitly, because an unratified legal correspondence is a liability if read as more than it is. It is a *proposed* correspondence between evidence artifacts and obligations, not a validated legal determination, and it has not been ratified by a regulator, auditor or legal scholar. Passing the axes is *necessary-not-sufficient* evidence: it demonstrates that a system produces reproducible traces, per-decision constraint logs, and calibrated-with-abstention behaviour, which are preconditions for the

Prior work	What it provides	What this protocol adds
Internal / assurance audit frameworks (Raji et al. 2020; Lam et al. 2024)	Organisational, architecture-agnostic evidence structure	Concrete per-decision tests with pre-declared pass criteria
Dataset-facing auditing, e.g. G-AUDIT (Drenkow et al. 2025)	Hypotheses about dataset label–attribute bias	A decision-facing verdict against a guideline-anchored rule
Conformal / selective risk control (Angelopoulos et al. 2024; Xu, Guo, and Wei 2025)	Finite-sample control of arbitrary losses under coverage	Instantiation for constraint *compliance* with a coverage floor closing abstention-gaming
Auditee-gaming of platform audits (Burnat and Davidson 2026b,a)	Temporal / metric gaming, audit-frequency floors	A per-decision deployment-coverage floor for clinical deferral
Confident learning (Northcutt, Jiang, and Chuang 2021)	Label-noise detection from model confidence	Surfaces the accuracy-vs-guideline gap as a deployed compliance failure, by patient group

Table S16: Positioning relative to prior art.

cited obligations. It does not constitute a legal determination of compliance, a certification of clinical safety, or a guarantee under future distribution shift. A hospital observing a full pass should read it as "the system emits the evidence an audit would need", not as "the system is compliant". Establishing the mapping's legal sufficiency through structured expert consultation is future work, and the empirical contribution of this paper does not depend on it.

S12 Neuro-Symbolic Framing

The protocol was developed with neuro-symbolic (NeSy) clinical systems as the motivating case, since that setting most sharply foregrounds the neural-to-symbolic interface the concept-semantics probe audits, and NeSy is frequently proposed as a route to trustworthy clinical AI on grounds of inspectability (d'Avila Garcez and Lamb 2023; Sheth, Roy, and Gaur 2023; Bhuyan et al. 2024), efficiency (Velasquez et al. 2025), the reasoning limits of scaled models (Belle and Marcus 2026) and safety in high-stakes applications (Bhattacharya and Ashish 2025). Domain systems build individual accountable deployments — for example a legally grounded NeSy framework for public-benefit eligibility (Sunny and Sivan-Sevilla 2026) — while evaluation work targets concept quality and reasoning shortcuts (Bortolotti et al. 2024; Marconato et al. 2023), and verification work provides formal guarantees for specific properties (Xie, Kersting, and Neider 2022; Ahmed et al. 2022). A recent review surveys robustness, uncertainty quantification and intervenability (Acharya and Song 2026).

Two points follow. First, the presence of symbolic structure does not confer accountability: the model audited in this paper *had* an explicit constraint attached and violated it, because the constraint was only logged and the training label was not guideline-faithful. That is the assumption — "it is neuro-symbolic, therefore trustworthy" — a falsifiable test exists to check. Second, and more importantly for this paper, nothing in the protocol requires a NeSy architecture. The three axes apply to any clinical model paired with an explicit, version-controlled rule set; NeSy is the motivating case, not a requirement, and the finding reported in the body is a fact about an administrative labelling convention, not about an architecture.

Axes we did not implement. Three axes are natural extensions but require artifacts our systems do not emit, and we name them rather than imply coverage. *Contestability* would test whether the system exports a recourse set robust to model updates (Jiang et al. 2024). *Rationale faithfulness* would subject symbolic traces to randomisation and sufficiency probes (Adebayo et al. 2018; Turpin et al. 2023). And a *probabilistic-logical* integration arm — registered as Semantic Probabilistic Layers (Ahmed et al. 2022) with DeepProbLog (Manhaeve et al. 2021) as the declared fallback — was not run (Appendix S1, row 6), so only two of three registered integration styles are evaluated: the pipeline-coupled arms in the body and the differentiable arm in Appendix S3.

S13 Material Moved from the Main Text for Length

The main text is held to eight pages. Nothing cut from it for length was deleted: each passage below is reproduced verbatim from the longer draft (section labels resolved to the main-paper section numbers), under the item number the body cites. Items are listed in the order a reader would meet them in the body; the `RESTORE-IF-REFS-EXCLUDED` list at the top of `main.tex` gives the priority in which they return to the body if the page limit excludes references. Item S13.10 is superseded: the body no longer asserts that the nadir-baseline and urine-output simplifications are conservative (main paper, Section 6).

S13.1 Full Data-and-Code-Availability and IRB statements (front matter)

Data and Code Availability This study uses two credentialed-access PhysioNet databases, MIMIC-IV (Johnson et al. 2023) and eICU (Pollard et al. 2018), which are available to approved researchers who complete PhysioNet credentialing and sign the applicable Data Use Agreements. Patient-level data, cohort tables, per-decision trace bundles, the clinician-adjudication packet and its unblinding key, and all other protected derivatives cannot be redistributed. The code repository linked from the main paper contains the audit harness, both cohort-construction pipelines, the frozen configuration holding every pass threshold, the analysis scripts (`03`–`13`), the pre-registration and its deviations log (`preregistration.md`, `preregistration_v14.md`), the full test suite, the claim-to-result trace table, and aggregate result JSONs; it carries no patient-level file, and a test fails if such a file becomes committable. The pre-registration tags (`prereg-v1.0` to `v1.3`) live in the authors' development repository and will be made public with the camera-ready version. Reproduction of the MIMIC-IV and eICU results requires independent credentialed PhysioNet access.

Institutional Review Board (IRB) This is a secondary analysis of two existing de-identified credentialed-access databases. MIMIC-IV was approved by the Institutional Review Boards of Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology, with waiver of informed consent for collection and release (Johnson et al. 2023); eICU was released under a similar approved arrangement (Pollard et al. 2018). Access was obtained through PhysioNet credentialing and database-specific Data Use Agreements. This is a secondary analysis of de-identified data and involved no interaction with human subjects.

S13.2 "What the task is, precisely" (Section 4)

What the task is, precisely. The label is an *administrative* AKI diagnosis (ICD-9 584.x / ICD-10 N17.x) recorded at the admission level, with no timestamp: it marks whether the admission *eventually* carried an AKI code, which in practice is a discharge-time billing determination. The features are drawn from the first 24 hours. The trained task is therefore retrospective prediction of an eventual administrative diagnosis from early data — not prospective early warning, and not detection of AKI present at the decision point. What we audit is the consequence *if such a score were read* as a decision aid about AKI now, which is how a deployed risk score is used; the constraint is evaluated against the contemporaneous first-24h creatinine trajectory, so the audit compares an eventual-diagnosis predictor against a present-state guideline rule. That asymmetry is the object of study, not an oversight, and it is why we describe conflicts as the model's high-confidence negative output disagreeing with a guideline-derived rule rather than as the model being wrong about AKI.

S13.3 Related Work, full version (Section 2)

Administrative labels imperfectly capture clinical phenotypes in AKI (Hwang et al. 2012; Ko, Venkatesan et al. 2018; Chanda et al. 2026; Høyer et al. 2026) and more broadly. Dataset-facing tools can surface label–attribute bias (Drenkow et al. 2025), and confident-learning methods can flag likely mislabeled rows (Northcutt, Jiang, and Chuang 2021). We ask a different question: not which rows are mislabeled, but whether a model trained on that label contradicts a clinical guideline on the decisions where the guideline speaks. A label-error rate need not reveal that concentration, and our headline is that it does.

Accountability frameworks such as model cards (Mitchell et al. 2019), internal audits (Raji et al. 2020), and assurance-style audits (Lam et al. 2024) describe the kinds of evidence systems should produce, but they do not provide a per-decision clinical conformity test. The statistical pieces are familiar: conformal risk control (Angelopoulos et al. 2024), selective risk control (Xu, Guo, and Wei 2025), calibration measurement (Guo et al. 2017), and known weaknesses of post-hoc explanations (Ribeiro, Singh, and Guestrin 2016; Lundberg and Lee 2017; Adebayo et al. 2018; Krishna et al. 2022; Slack et al. 2020). Pairing a predictor with explicit, version-controlled rules is the alternative audited here (Lu et al. 2024). Our contribution is to assemble these ideas into a pre-registered, decision-facing audit with negative controls, separate axis verdicts, and a coverage floor that prevents a model from passing simply by deferring the high-risk subgroup.

S13.4 T3 binning disclosure (Section 3)

T3 measures Expected Calibration Error (Guo et al. 2017) coupled to abstention below a confidence threshold τ, with 10 equal-width bins on top-label confidence as pre-registered — of which only five are reachable, since that confidence is bounded below by 0.5. Coarser bins understate ECE and so bias the statistic *towards* the T3 pass, in favour of the models we audit; we disclose this rather than re-bin a pre-registered estimator, noting the margins are wide (CI upper 0.004 against a 0.10 bound).

S13.5 "What this control does and does not show" (Section 5.5)

What this control does and does not show. This is a positive control, not a causal centrepiece. On the constraint-applicable subgroup the KDIGO target is already positive for 95.4% of cases against 37.0% for the ICD target, so about 81% of the violation-rate drop follows from target arithmetic rather than learned behaviour. The control establishes that the constraint is triggerable, the harness responds, and the failure is a property of the target rather than of model architecture; it does not prove that any guideline-faithful target would solve deployment compliance.

S13.6 Within-eICU replication paragraph (Section 5.5)

Independent multicentre replication on eICU. MIMIC-IV is the primary patient-level evaluation; eICU is independent

multicentre replication across 205 hospitals under a different administrative labelling system (Section S8). On eICU the model is again accurate (AUROC 0.852) and calibrated, denies AKI on a larger share of deployed severe cases (0.93 vs 0.71, on 4,142 acute-severe stays), and passes neither policy's certificate. The failure surface differs, and we report it: the eICU model abstains far more on the severe subgroup, so the logged policy deploys on only 39% of severe stays (5th percentile 0.34) and fails the certificate on *both* limbs, coverage and violation, while the enforced policy covers 3%. Label agreement is in the same regime (43% vs 37%), and the non-monotonic creatinine probe replicates (Figure S3). The eICU arm does not support a patient-level claim because it groups by hospital stay rather than patient; that limitation is kept explicit.

S13.7 Three figures for the same quantity (Section 5.1)

Three figures describe the same quantity in this paper and are labelled where they appear: the across-seed mean (0.714), the 95th percentile the certificate tests (0.744), and the mean on the matched cohort intersection of Section 5.5 (0.720).

S13.8 Gaming and governance, full paragraph (Section 6)

Gaming and governance. An audit a deployer can pass without reducing harm is worthless; our enforced policy exposed one such move — zero violations by mass abstention — which the coverage floor closes. Each axis produces evidence of the kind transparency and risk-management instruments call for (European Parliament and Council 2024; U.S. Food and Drug Administration 2025; National Institute of Standards and Technology 2023), but the mapping is proposed and not regulator ratified (Section S11).

S13.9 Constraint-blind control sentence (Section 5.5)

The constraint-blind control also shows why T2 cannot be read as concept understanding: without creatinine it defers more often and commits fewer hard violations, even though it cannot represent the rule-relevant quantity.

S13.10 "Clinical caveats" paragraph (Section 4; superseded wording)

Clinical caveats. Three simplifications bound interpretation and are conservative with respect to the central claim: baseline creatinine is the within-window nadir rather than a pre-admission value; only the serum-creatinine limb is implemented, so urine-output AKI is under-counted; and community- and hospital-acquired AKI are not separated. The constraint is our version-pinned operationalisation of KDIGO 2012 criteria, not a clinician-ratified phenotype; this limitation is tested by the released adjudication protocol and revisited in Section 6.

S13.11 Concept-semantics probe and negative controls, full paragraph (Section 3)

The concept-semantics probe, and negative controls. T2 is *syntactic*: it verifies that outputs respect the rules, not that a decision depends on the constrained concept. We therefore sweep the constraint-relevant feature counterfactually and record the induced change in predicted risk; on a small tabular task (Section S2) it separates a constraint-sensitive model from one trained without the constrained feature. A suite that cannot fail is not evidence, so four negative controls are registered with their required outcomes, and failures to fire count against the protocol. Section S1 contains the complete deviations log, including registered items we did not deliver.

S13.12 "The conditional in our claim" paragraph (Section 1)

The conditional in our claim is deliberate, so we state its limit: we have no evidence that a named deployed system is trained on ICD AKI codes and do not assert it. What makes the conditional worth auditing is that both halves are documented independently — the convention is real in the literature, and deployed AKI alerting has already been shown capable of harm (above). Closing it needs a label-provenance audit of a fielded system, the natural next study rather than this one.

S14 Full Tables for the Three New Evidence Blocks

The body (Sections 5.4 and 5.6) reports the restricted-cohort re-audit, the held-out-hospital certificate and the clinician adjudication in compact form. The complete tables follow. All three are post hoc analyses added after the pre-registered study (not pre-registered) and changed no registered threshold; the deviations log (Section S1; v1.12–v1.13 in `preregistration_v14.md`) records them. Every figure is read from the named results key; `claim_verification_w3phiai.csv` lists each with its key path.

Restricted cohort (model statistic). Table S17 gives the per-stratum and pooled certificate; Table S18 the patient-clustered intervals. The pooled violation rate weights strata by deployed decisions (the rate's denominator); weighting by severe cases instead moves it by 0.0004.

Held-out hospitals. Table S19 gives both policies' certificates on the 103 held-out eICU hospitals. The failing limb is coverage, as in the within-eICU certificate of Section S8 (coverage 0.39); the violation cap is breached as well. Seeds vary the patient-level training split inside the training hospitals, hence the model; they do not vary the hospital partition, so the quantiles describe model variability on a fixed held-out set.

Clinician adjudication. Table S20 is the per-cell table; Tables S21 and S22 are the post-hoc cross-tabulation. Two bounds apply (main paper, Section 5.6): every decided verdict coincides with the KDIGO stage$\geq$1 creatinine criterion, so the exercise validates the operationalisation against a

Stratum	Severe/split	min n	Coverage	Violation rate (p5–p95)	ICD agr.	Status
Prior renal replacement	716	627	0.890	0.941 (0.918–0.956)	0.079	OK
Observable history, no replacement	528	499	0.731	0.479 (0.449–0.510)	0.597	UNDETERMINED
No prior admission (unobservable)	529	499	0.743	0.579 (0.527–0.617)	0.557	UNDETERMINED
Surviving strata, pooled	1056	1006	0.737	0.529 (0.492–0.563)	—	determinate

Table S17: Model-level restricted-cohort audit by renal-history stratum, logged policy, 20 splits (means across splits; violation-rate 5th–95th percentiles in parentheses). The three strata partition all severe cases. Pooled surviving cohort: 95th-percentile violation 0.563 against cap 0.05 (FAIL), 5th-percentile coverage 0.711 against floor 0.50 (pass); verdict FAIL on the violation limb. Per-stratum status follows the registered $n \geq 500$ floor (one split each at $n = 499$ in the two surviving strata). Source: `results/violation_rate_by_renal_history.json`.

	Per split (median of 20)		Pooled (descriptive)	
Statistic	point	95% CI	point	95% CI
Deployed violation rate	0.528	0.492–0.566	0.529	0.509–0.548
Coverage on severe cases	0.737	0.710–0.765	0.737	0.723–0.750
ICD-label agreement on severe cases	0.574	0.542–0.607	0.577	0.561–0.595

Table S18: Patient-clustered 95% confidence intervals on the restricted (surviving) cohort: `subject_id`-clustered nonparametric bootstrap, $B = 1000$, seed 0. Left: computed per split on that split's test subjects, median across the 20 splits of point, lower and upper bound. Right: one interval over the stacked test decisions of all 20 splits (21,129 decisions, 3,163 subjects, of whom 3,148 appear in more than one split's test set), which is descriptive of the stack and not an inferential interval. Added in v1.13; no threshold changed.

nephrologist's reading of the guideline rather than establishing clinical AKI independently; and the 38 blanks are not random, concentrating in rule-severe cases (30 of 38) and in the elevated-at-arrival trajectory class (27 of 38). Inter-rater reliability is not computable with one adjudicator.

	Coverage on severe			Deployed violation rate			
Policy	mean	p5	p95	mean	p5	p95	**Verdict**
Logged	0.307	0.196	0.369	0.973	0.940	0.995	FAIL (coverage)
Enforced	0.009	0.002	0.023	0.000	0.000	0.000	FAIL (coverage)
Floor / cap		≥ 0.50				≤ 0.05	

Table S19: Held-out-hospital certificate on eICU: trained on 102 hospitals (92,174 stays), every decision evaluated at the other 103 (77,651 stays, 1,873 guideline-severe), 20 seeds varying the model, not the partition (partition RNG seed 0). Held-out AUROC 0.849 (range 0.826–0.854). Per site: 80 held-out hospitals have severe stays; 0 reach the registered $n \geq 500$ floor, so every per-site certificate is UNDETERMINED; among the 33 sites with at least 20 severe stays per seed, deployed violation rates run 0.846–1.000 (median 0.988). Post hoc, added after the pre-registered study; not pre-registered. Source: `eicu_pipeline/results/heldout_hospital_certificate.json`.

Cell	**Cases**	**Decided**	**Blank**	**Agree**	**vs rule**	**vs code**
A: severe, coded	40	27	13	27	1.000	1.000
B: severe, uncoded (headline)	80	63	17	63	1.000	0.000
C: near threshold	60	52	8	40	0.769	0.635
D: clearly not severe	20	20	0	19	0.950	0.800

Table S20: Clinician adjudication, per sampling cell: one nephrologist, blind to label, model output and cell; 200 cases, 162 decided, 38 blank ("cannot tell", excluded from the denominator, never scored as "no"). "Agree" and "vs rule" are agreement with our severity rule on decided cases; "vs code" is agreement with the administrative label. Pre-declared thresholds: cell-B agreement ≥ 0.7, overall $\kappa \geq 0.6$; observed $\kappa = 0.835$ against the rule and 0.023 against the code. Pooled figures are reported only beside this table because the cell mix was chosen by us. Source: `results/clinician_adjudication.json`, `results/adjudication_by_stratum.json`.

Stratum	**Cases**	**Decided**	**Blank**	**Agree**	**Agreement**
Renal history (block-H partition)					
Prior renal replacement	63	53	10	51	0.962
Observable history, no replacement	72	56	16	52	0.929
No prior admission	65	53	12	46	0.868
Creatinine trajectory (error-taxonomy axis A)					
A1: elevated at arrival (≥ 4.0 nadir)	92	65	27	65	1.000
A2: rise from near-normal (< 2.0)	25	25	0	22	0.880
A3: rise from elevated base (2.0–4.0)	83	72	11	62	0.861

Table S21: Adjudication verdicts cross-tabulated against the renal-history partition of the restricted-cohort analysis and against the creatinine trajectory classes (post hoc, v1.13). Agreement is with our rule on decided cases. The blanks concentrate in A1 (already elevated at arrival); every decided A1 case agrees with the rule. Source: `results/adjudication_by_stratum.json`.

Cell	Stratum	Cases	Decided	Agree	Agreement
Cell × renal history					
A	prior RRT	4	3	3	1.000*
A	no prior RRT	17	10	10	1.000
A	unobservable	19	14	14	1.000
B	prior RRT	45	37	37	1.000
B	no prior RRT	14	10	10	1.000
B	unobservable	21	16	16	1.000
C	prior RRT	14	13	11	0.846
C	no prior RRT	30	25	22	0.880
C	unobservable	16	14	7	0.500
D	no prior RRT	11	11	10	0.909
D	unobservable	9	9	9	1.000*
Cell × trajectory					
A	A1	30	19	19	1.000
A	A2	1	1	1	1.000*
A	A3	9	7	7	1.000*
B	A1	62	46	46	1.000
B	A2	2	2	2	1.000*
B	A3	16	15	15	1.000
C	A2	2	2	0	0.000*
C	A3	58	50	40	0.800
D	A2	20	20	19	0.950

Table S22: Joint cells of the adjudication cross-tabulation (non-empty combinations only). *Fewer than 10 decided cases; 6 such cells are listed under `small_cell_warning` and should not be read as rates. Ten of the twelve near-threshold (cell C) disagreements fall in trajectory class A3. Source: `results/adjudication_by_stratum.json`.

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